



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 23rd

Date: 14/12/2025

Mid-Term Examination

Semester: Winter 2025

Time: 1 and 1/2 hour

Total Marks: 30

Answer any **five** questions from the following.

5x6= 30

- 1 (a) Explain Artificial intelligence and its use cases. 2
(b) Define Machine Learning and its different types. 2
(c) When should we use statistics, machine learning and deep learning algorithms? 2
- 2 (a) Sigmoid function is defined as: $S = \frac{1}{1+e^{-x}}$. Write a sigmoid function using python. 2
(b) Given that a quadratic equation $ax^2 + bx + c = 0$. Write a python function that solve this given problem using the following solution $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ 2
(c) Given that, $(x - h)^2 + (y - k)^2 - r^2 = 0$. Write a python function for calculating the area of that given problem. 2
- 3 (a) How to measure central tendency of a given list like data. Compare all of them and chose the best one. 2
(b) Given that, $\vec{v} = [e_1, e_2, e_3, e_4, e_5]$ find average and mode using python. 2
(c) Given that, $\vec{v} = [e_1, e_2, e_3, e_4, e_5]$, Find the variance and standard deviation using python function 2
- 4 (a) Given that $\vec{v} = [1, 2, 2]$ 3
1. Calculate the vector length.
2. Compute the length of $m \cdot \vec{v}$, where $m=0$
3. Compute $|\vec{v}|$
(b) Given that $\vec{v} = [2, 2, 2]$, Compute angles i.e. α, β, γ 2
(c) Given that $|a||b| \cos \theta = 0$. 1
Are they orthogonal?
- 5 (a) Given that $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 7 & 8 & 9 \end{bmatrix}$ 2
1. Find the $\det(A)$.
2. Find A^{-1}
(b) Given that $f(x) = x^2 + 5x + 1$ and $g(x) = x^3 + 2x + 1$ 2
1. Calculate: $\frac{d}{dx}(f(x) + g(x))$
2. Calculate: $\int (f(x) - g(x)) dx$
(c) Given that, a coin toss data is [H,T,H,T,H,T] 2
What will be the joint probability?

6 (a) Given that a dataset of weather,

Day	Sky Condition	Rain
1	Sunny	No
2	Cloudy	No
3	Rainy	Yes
4	Sunny	No
5	Cloudy	Yes
6	Rainy	Yes
7	Sunny	No
8	Cloudy	No
9	Rainy	Yes
10	Sunny	No

Calculate the probability of rain, using Bayes theorem.

(b) Given that, Contingency table

Gender	Pass	Fail	Total
Male	6	5	11
Female	5	5	10
Total	11	10	21

H_0 = Gender is related to pass or fail. Test that hypothesis and conclude.

(c) True or false ? "Probability can be negative"

7 a) True or False?

1. Machine learning is not a branch of AI but Statics.
2. Supervised algorithm uses "Labeled" data.
3. Python has the best Eco system for AI.
4. Deep Learning is better for Computer Vision

(b) 1. Correlation range is -1 to 1

2. IQR represent Q1 to Q3
3. Outliers are in Q1 and Q4
4. Skewness measures the asymmetry of dataset

(c) 1. Average can not be influenced by outliers.

2. Alternative hypothesis is True when Null hypothesis is False.

1-b machine learning is a subis of AI allows to automatically learn and improves from experience without being explicitly programmed. It us algorithms and statistical models to analyze and learn patters from data making decisions.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE471: Computer and Network Security

Batch: 23rd
Date: 17/12/2025

Mid-Term Examination

Semester: Fall 2025
Time: 1 Hour 30 minutes
Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

1. a) State the differences between active and passive attack. 2
b) Suppose Alice and Bob want to use symmetric key encryption for secure communication. They want to establish their secret key using Diffie-Hellman Key Exchange Algorithm. Show the complete key establishment process if $g = 11$, $n = 19$, $x = 5$ and $y = 4$. 4
2. a) Explain the differences between decoding and decryption. 2
b) What are the limitations of classical encryption algorithms? 2
c) Describe the key management problem in symmetric key encryption. 2
3. a) Let $p = 7$, $q = 13$ and $e = 5$. Calculate the private and public key using RSA algorithm. 2
b) Suppose message $M = 10$, show the complete encryption and decryption process using the keys you have just calculated. 4
4. a) What is digital signature? 1
b) How can an attacker perform a man-in-the-middle (MITM) attack in an unencrypted communication environment with ensuring the attack remains completely unnoticeable by the authorized users? Explain the process in detail. 2
c) The bold text is encrypted using Shift Cipher. Show the cryptanalysis and determine the plaintext. Also determine the key that was used for encryption. 3

SJB DJFW SJB RJ
5. a) State the six key characteristics of Hashing. 2
b) Explain how encryption is used to achieve authentication with appropriate illustrations. 4
6. a) What is brute force attack? 1
b) Explain the CIA triad. 2
c) Explain how a Message Authentication Code (MAC) provides authentication with the help of encryption. Provide necessary Illustration. 3
7. a) What is rubber ducky? Explain how social engineering is important to perform attack using rubber ducky. 2
b) Show the encryption process of the following message and determine the ciphertext 'C' using Row Column Cipher where row = 4, column = 4 and key = 3124. 2

UNCHARACTERISTIC

c) "MD5 is a broken hashing algorithm" – justify the statement. 2

Answer any five questions.

Marks: 5 x 6 = 30

1. (a) Write down the differences between Compiler and Interpreter. 3
 (b) What is DFA? Let $\Sigma = \{a, b\}$ and $L = \{abb\}$, Design a DFA for L. 3

2. (a) Write down the Role of a Parser with diagram. 2
 (b) What are Alphabet and String? Take an example of a string and find the length of that string. 2
 (c) $S \rightarrow AB$ 2
 $A \rightarrow c/aA$
 $B \rightarrow d/bB$
 From above grammar draw the parse tree and create a word: **acbd**

3. (a) What is an ambiguous grammar? 2
 (b) Draw the diagram of the Language processing system. 2
 (c) Draw an NFA from the Regular Expression: $(a^+ \cup aab)^+ bbab$. 2

4. (a) Take an expression for example and show how the Syntax analyzer and the Intermediate code generator work. 3
 (b) Convert the following NFA to DFA: $L = \{3^{rd} \text{ symbol from the LHS is 'b'}\}$. 3

5. (a) $E \rightarrow TE'$ 3
 $E' \rightarrow *TE' \mid \epsilon$
 $T \rightarrow FT'$
 $T' \rightarrow \epsilon \mid +FT'$
 $F \rightarrow id \mid (E$
 Find out FIRST() and FOLLOW() from this grammar.

- (b) $S \rightarrow iCtSS_1 \mid a$ 3
 $S_1 \rightarrow \epsilon S \mid \epsilon$
 $C \rightarrow b$
 Explain whether this grammar will be accepted by LL(1) parser or not.

6. Describe briefly all phases of compiler with example. 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 415: Communication Engineering

Batch : 23rd

Date: 09/12/2025

Mid-Term Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks : 30

Answer any five questions:

5 × 6 = 30

1. (a) Why does a router normally have more than one interface? Explain. 3
(b) How many IP addresses and how many link-layer addresses should a router have if it is connected to five links? Explain. 3
2. (a) Given the Dataword **101001111** and the divisor **10111**, show the generation of the CRC Codeword at the sender site (using binary division). 3
(b) Create a table for Cyclic Redundancy Check (CRC), where $K = 4$ and $N = 7$. 3
3. (a) Draw and explain the FSM for the Stop-and-Wait Protocol. 3
(b) Using the following specifications, draw a finite state machine with three states (I, II, and III), five events, and six actions: 3
 - i. If the machine is in state I, two events can occur. If event 1 occurs, the machine moves to state II. If event 2 occurs, the machine performs actions 1 and 2 and moves to state III.
 - ii. If the machine is in state II, two events can occur. If event 3 occurs, the machine remains in state II. If event 4 occurs, the machine moves to state III.
 - iii. If the machine is in state III, three events can occur. If event 2 occurs, the machine remains in state III. If event 3 occurs, the machine performs actions 1, 2, 4, and 5 moves to state II. If event 5 occurs, the machine performs actions 1, 2, and 6 and moves to state I.
4. (a) Explain the Control field format for different HDLC frame types. 3
(b) Compare and contrast byte-oriented and bit-oriented protocols. 3
5. (a) Draw a table for the Link Control Protocol packet. 3
(b) In PPP, we normally talk about user and system instead of sending and receiving nodes; explain the reason. 3
6. (a) Why is the throughput of Slotted ALOHA 2 times greater than Pure ALOHA? Explain. 3
(b) Sketch the flow diagram for three the persistence methods. 3
7. (a) Illustrate an Ethernet frame. 3
(b) Illustrate common Gigabit Ethernet implementations. 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE413: Computer Graphics

Batch: 23rd

Date: 07/12/2025

Mid Term Examination

Semester: Fall 2025

Time: 1 hour and 30 minutes

Total Marks: 30

Answer any five questions from the following:

5x6= 30

- 1 (a) Explain any four major applications of Computer Graphics in design and simulation. 3
(b) List the components of an Interactive Computer Graphics system and briefly describe any three of them. 3
- 2 (a) Describe the basic structure and purpose of a Cathode Ray Tube (CRT) with a neat labeled diagram. 3
(b) State the advantages and disadvantages of vector graphics. 3
- 3 (a) Define Computer Graphics. 2
(b) Differentiate between the DDA and Bresenham's Line Drawing Algorithms. 4
- 4 (a) Using the DDA algorithm, compute the pixel coordinates for the line from (5, 4) to (13, 10). 3
(b) Using the Mid-Point Line Algorithm, compute the first five plotted points for the line joining (4, 6) and (12, 11). 3
- 5 (a) Explain the concept of 8-fold symmetry in circle drawing. 2
(b) Apply the Mid-Point Circle Algorithm to plot the circle with center (2, -3) and radius 6. Show all symmetric points for the first octant. 4
- 6 (a) Write the algorithmic steps for the Bresenham's Line Drawing Algorithm. 3
(b) Rasterize the line between (7, 2) and (14, 9) using Bresenham's algorithm. Show all decision variable updates. 3
- 7 (a) Explain the difference between object transformation and coordinate transformation in Computer Graphics. 3
(b) Apply the following transformations to the point B(4,3,1): 3
 - i. Scale the point with $S_x = 0.5$, $S_y = 2$, $S_z = 1$.
 - ii. Rotate the point 180° around the y-axis.
 - iii. Translate the result by $(T_x, T_y, T_z) = (1, 1, -2)$.